METRIC

DOD-W-24644(SH)
11 September 1984

MILITARY SPECIFICATION

WELDING MACHINE, ARC, DIRECT-CURRENT, 300 AMPERE, VARIABLE VOLTAGE, TRANSFORMER-RECTIFIER TYPE, DAMAGE CONTROL, LIGHT WEIGHT, PORTABLE (METRIC)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable welding machines of the direct current, transformer-rectifier design for shipboard use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-E-2036	- Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-22200/1	- Electrodes, Welding, Mineral Covered, Iron-Powder, Low-Hydrogen Medium and High Tensile Steel, as Welded or Stress-Relieved Weld Application.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 5523, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3431

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-756 - Reliability Modeling and Prediction.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric).
MIL-STD-1600 - Reliability Prediction of Monolithic Integrated Circuits.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-472 - Maintainability Prediction.

2.1.2 Other Government documents and publications. The following Government document and publication forms a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR

Code of Federal Regulations, Title 29
Part 1910 - Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0967-LP-319-5010 - Electronic Equipment Reliability, Handbook for the Prediction of Shipboard and Shore.

(Copies of specifications, standards, handbooks and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Safety. The welding machine shall provide fail-safe features for safety of personnel during the installation, operation, maintenance, and repair or during interchanging of a complete welding machine assembly or component parts. Welding machine design for personnel safety shall be equal to or better than the appropriate requirements of the Occupational Safety and Health Administration (OSHA) as identified in Title 29, Part 1910, of the Code of Federal Regulations. Except as required herein, design for safety shall be in accordance with MIL-E-917.

3.2.1 Accidental contact. The design of the welding machine shall provide protection to personnel against accidental contact with voltages in excess of 30 volts root mean square (Vrms) alternating current (a.c.) or direct current (d.c.) except between the electrode holder and the work piece.

3.3 Materials, methods, parts and processes. Unless otherwise specified herein, commercial materials, methods, parts and processes may be used, provided that failure to meet the performance or environmental requirements of this specification shall not result from the use of such materials, methods, parts and processes.

3.3.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

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3.3.2 Prohibited materials. Materials prohibited by MIL-E-917 shall not be used.

3.3.3 Hazardous items and toxic materials. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred to the Naval Sea Systems Command. Regardless of any other requirements, materials and parts containing asbestos, cadmium, lithium, mercury, radioactive material or materials giving off toxic fumes under operating or casualty conditions shall not be used, except that sheet metal screws and assembly bolts with cadmium protective coating may be used.

3.3.4 Material compatibility. Materials which through outgassing or other physical phenomena cause deterioration of other materials or degradation of welding machine performance shall not be used.

3.4 Design and construction.

3.4.1 Composition. The welding machine shall be completely assembled, portable, and shall comprise all equipment necessary for furnishing electric energy to the welding circuit, except the input power supply cable (see 3.4.3.1).

3.4.2 Construction. The welding machine enclosure shall be in accordance with the class 2, dripproof type for electric equipment requirements of MIL-E-2036, except that the angle of incidence shall be limited to 15 degrees.

3.4.2.1 Lifting means. The welding machine shall be provided with handles or hand rails suitable for lifting the machine.

3.4.3 Power input. (See 3.7.1).

3.4.3.1 Input power supply cable. When required, the input power supply cable shall be specified by the contracting activity (see 6.2.1).

3.4.3.2 Switch or circuit breaker. The primary or input circuit to the welding machine shall contain a manually or magnetically operated circuit breaker or manually operated switch which shall be so connected as to open all input power circuits to the welding machine.

3.4.3.3 Output terminal connections. Welding terminals shall be of the stud and wing nut or hex nut type. Terminals and fittings shall be copper, brass or bronze of ample conducting capacity for rated current. Welding terminals shall be provided with an identification plate to show the connections for operation of the machine with straight (electrode negative) or reverse (electrode positive) polarity.

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3.4.4 Welding control. The welding control shall include contact making devices, rheostats, switches and other equipment necessary to provide adjustments for the range of current and voltage specified in 3.7.3.2.3. Welding controls shall be mounted in a dripproof protected enclosure as specified in 3.4.2. Welding controls shall be easily accessible and shall be of such design that they can be readily operated by an operator wearing a welder's glove. Welding terminals shall be protected from accidental contact or short circuit. Other live parts shall be contained within the welding control enclosure. Adjusting devices shall be provided with calibrated plates designed to show the magnitude of current for each setting.

3.4.5 Insulation.

3.4.5.1 Insulation resistance. The insulation resistance of all circuits shall be not less than 10 megohms when measured cold (test voltages which are greater than the maximum rated voltage of components shall not be applied across such components during the inspection tests).

3.4.5.1.1 Design and construction. Design and construction of welding machine components shall be such that a 500 volt d.c. insulation resistance measurement in accordance with 4.6.2.2 shall not damage any components under the following conditions:

(a) Output terminals connected together.
(b) Ohmmeter connected between those terminals and ground.
(c) No insulation failure exists, that is, insulation resistance is above 10 megohms.
(d) Input terminals connected together.

3.4.5.2 Dielectric strength. The welding machine shall be so designed and constructed as to withstand for a period of 1-minute a dielectric test voltage to ground of 1000 Vrms plus twice rated. The frequency shall be 60 Hz and the waveshape shall be sinusoidal. Voltages greater than the maximum rated voltage of components shall not be applied across these components during the inspection tests (see 4.6.2.8).

3.4.5.3 Class of insulation. Insulation materials shall be class B, F, H or N, as specified in MIL-E-917, and shall be suitable to permit continuous and intermittent operation of the welding machine in the performance of all functions specified (see 3.7), under the environmental conditions specified (see 3.8), without causing damage to or degrading effects in the insulation materials.

3.5 Reliability. The mean time between failure (MTBF) for the welding machine specified herein shall be 5000 hours for operation under the conditions of temperature and humidity ranges as specified in 3.8, and under the performance requirements as specified in 3.7.

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3.5.1 Reliability prediction for electronic components. When specified in the contract or order (see 6.2.1), a reliability prediction shall be performed in accordance with the procedure of type II of MIL-STD-756. MIL-HDBK-217 shall be used as the basic data source, supplemented where necessary in accordance with MIL-STD-1600 and NAVSEA 0967-LP-319-5010. The prediction shall be based on a detailed thermal/electrical/mechanical stress analysis of each part and shall reflect the part derating requirement as specified in 3.5.2 (see 6.2.2).

3.5.2 Derating. Derating of electronic parts and materials used shall be accomplished in accordance with requirement 18 of MIL-STD-454, based on technical guidance as specified in MIL-HDBK-217.

3.6 Maintainability. Unless otherwise specified (see 6.2.1), the welding machine shall have an on board (ship, platform, submarine) mean-time-to-repair (MTTR) no greater than 60 minutes, and maximum repair time (M-max) of 120 minutes at the 95th percentile. The time to repair shall include diagnosis, replacement and verification of corrective action.

3.7 Performance. The welding machine provided under this specification shall perform all of the functions specified herein, within the interface constraints of input power specified in 3.4.3.1, and environmental conditions specified in 3.8.

3.7.1 Power input interface.

3.7.1.1 Input current waveform. The operation of the welding machine shall have the minimum harmonic distortion effect on the electric system. The operation of the welding machine of the following specified rating shall not cause single harmonic line currents to be generated that are greater than 3 percent of the unit's full rated load fundamental current between the 2nd and 32nd harmonic:

Frequency of power source (Hz)	Rating of unit
60	1 kilovoltampere (kVA) or more

Additionally, currents with frequencies from the 32nd harmonic through 20 kilohertz (kHz) shall not exceed 100/n percent of the unit's rated full load fundamental current, where n is the harmonic multiple number. Units with power ratings less than those specified above shall be current amplitude limited such that no individual harmonic line current from the 2nd harmonic through 20 kHz exceeds a magnitude of 100/n percent of the unit's rated full load fundamental current.

3.7.1.2 Voltage and frequency tolerance. The nominal voltage shall be 440 volts. The user voltage tolerance shall be 440 volts $\pm$ 5 percent. The nominal frequency tolerance shall be 60 Hz. The user frequency tolerance shall be 60 Hz $\pm$ 3 percent. The welding machine shall operate satisfactorily within this tolerance and have a temperature rise not to exceed that specified in 3.7.6 when tested in accordance with 4.6.2.3.2.

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3.7.1.3 Voltage and frequency transient tolerance and recovery. The welding machine performance shall be evaluated under the following conditions of transient voltage, transient frequency and recovery time:

Condition	Transient voltage	Transient frequency
Upper limit	440 plus 20 percent	60 Hz plus 5-1/2 percent
Lower limit	440 minus 20 percent	60 Hz minus 5-1/2 percent
Recovery time	2 seconds	2 seconds

3.7.1.4 Voltage spike. The welding machine shall be evaluated for its capability to withstand the voltage spike as shown on figure 1 when tested in accordance with 4.6.2.3.4.

3.7.2 Suitability of operation. The welding machine shall be suitable for welding with MIL-7018 electrodes in accordance with MIL-E-22200/1. The characteristics shall be such that an operator qualified in accordance with MIL-STD-248 may strike and maintain a stable arc and satisfactorily weld at any setting (see 6.4.6) within the specified range of the equipment under normal conditions without losing the arc more than once per minute.

3.7.3 Current.

3.7.3.1 No load current. The maximum input line current with no output load shall not exceed 2.0 amperes.

3.7.3.2 Output load current.

3.7.3.2.1 Rated current. The rated load current of the welding machine shall be as specified in table I.

TABLE I. Size and ratings[1/].

Rating of equipment		Minimum range of output load current (from 20 volts to rated voltage)
Rated load current (amperes)	Rated load voltage (volts)	(amperes)
300	32	30 to 375

1/ Ratings shall be based on continuous welding at rated load current of 300 amperes at a 100 percent duty cycle (see 6.4.3), except that when equipment is operating at 375 amperes the duty cycle shall be 50 percent.

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3.7.3.2.2 Nominal arc voltage and current. Nominal arc voltage shall be 20 plus 0.04 times load current (E = 20 + 0.04 I).

3.7.3.2.3 Current adjustment. Current within the range specified in table I shall be obtained at any voltage between 20 volts and rated voltage inclusive. There shall be no setting at which it will be difficult to weld. Current shall be adjustable between the minimum and maximum values specified in table I in not less than 30 increments. Not more than two controls shall be used for current adjustment. At rated voltage, the increase in current from any setting to the next higher setting shall not exceed 5 percent of the current at the lower setting.

3.7.3.2.4 Current stability. The welding machine design shall be such that for a given current setting at a given ambient temperature, the current shall vary not more than plus or minus 5 percent from the initial current.

3.7.4 Output.

3.7.4.1 Rated voltage. The rated voltage of the welding machine shall be as specified in table I.

3.7.4.2 Open-circuit voltage. The maximum open-circuit d.c. voltage (see 6.4.5), shall not exceed 80 Vrms for welders with more than 10 percent ripple voltage, or 100 volts average for equipment having ripple voltage of 10 percent or less.

3.7.4.3 Volt-ampere characteristics. Volt-ampere characteristic curves (see 6.4.9) shall show that for a given current setting the voltage decreases at all points with an increase in current. No point of a curve shall indicate a current higher than that for short-circuit.

3.7.5 Rated output. The rated output of the welding machine shall be established as that obtained with the welding machine delivering rated load amperes at rated load voltage in accordance with table I and operating at rated input voltage and frequency.

3.7.5.1 Duty cycle. The welding machine shall operate satisfactorily with a 100 percent duty cycle at rated output load amperes and volts.

3.7.6 Temperature rises. Temperature rises shall not exceed allowable temperature rises as specified in MIL-E-917 when the welding machine is operating with a 100 percent duty cycle at rated load amperes and volts.

3.7.7 Efficiency. The efficiency of the welding machine shall not be less than the applicable values shown in table II and shall be determined in the manner as specified in 4.6.2.9.

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TABLE II. Efficiency values.

Percent of rated output	300 ampere equipment percent efficiency
25	60
50	65
75	70
100	75
125	70

3.7.8 Electromagnetic interference requirements. The welding machine shall conform to the radiated interference limits for class C3 (commercial electrical) equipment and the UM05 requirement for conducted and radiated broad band electromagnetic emissions in accordance with the emission and susceptibility requirements and limits as specified in MIL-STD-461. These requirements apply for no load and loaded over the rated current range when operating into a load resistor. The requirements are not applicable for welding conditions.

3.8 Environmental. The welding machine shall operate under the following environmental conditions specified in 3.8.1 through 3.8.5.

3.8.1 Temperature.

3.8.1.1 Operating temperature. The welding machine shall meet the performance requirements as specified in 3.7 when operated in any ambient temperature within the range of table III.

TABLE III. Environmental conditions.

Temperature operating	Humidity operating
0 to 40°C	10 to 95 percent R.H.

3.8.1.2 Non-operating temperature. Storage at any temperature within the temperature range of minus 40 to 71°C shall not damage the welding machine. After the temperature of the welding machine is stabilized to temperatures within the operating temperature range following storage, the welding machine shall meet the requirements as specified in 3.8.1.1.

3.8.2 Humidity. The welding machine shall meet the requirements as specified in 3.7 when exposed to an ambient humidity within the humidity range of table III for both continuous and intermittent periods.

3.8.3 Vibration. The welding machine shall meet the requirements as specified in 3.7 and the type I vibration requirements of MIL-STD-167-1.

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3.8.4 Shock. The welding machine shall meet the grade A, light weight type A requirements as specified in MIL-S-901.

3.8.5 Inclination. The welding machine shall meet the requirements as specified in 3.7 when permanently or temporarily inclined at any angle up to 15 degrees from the normal operating position.

3.9 Dimensions and weight. The overall dimensions shall be such as to allow the welding machine to pass through a circular hatch, 635 millimeters (mm) (25 inches) in diameter and through a hatch 508 mm (20 inches) wide by 965 mm (38 inches) high having four rounded corners with radii of 254 mm (10 inches). The weight shall not exceed 45.4 kilograms (kg) (100 pounds), without the input power supply cable and the welding leads attached.

3.10 Color. The welding machine shall be painted grey in accordance with MIL-E-917.

3.11 Identification, marking and labeling.

3.11.1 Labeling of hazardous items or toxic material. Any hazardous item or toxic material shall be identified. Any equipment containing hazardous items or toxic material shall be labeled in accordance with the requirements of public law including the Federal Hazardous Substance Labeling Act. In addition, the label shall include as applicable: name of product; quantity; warning symbol; signal word designating degree of hazard; affirmative statement of hazards; precautionary measures covering actions to be followed or avoided; instruction in case of contact or exposure; antidotes and notes to physicians; instruction in case of fire, spill, or leak; instructions for handling and storage; and disposal instructions. Characteristics and operating hazards which require labeling include: toxic, highly toxic, irritant, corrosive, strong sensitizer, combustible liquid, flammable, extremely flammable liquid, dangerously reactive, pressure generating, explosive, magnetic, ionizing radiation, non-ionizing radiation, high voltage, implosion, noise, and etiologic agent.

3.11.2 Identification plate marking. The data marked on main identification plates shall include the following items:

(a) Manufacturer's name, identification symbols, serial number and contract number.
(b) Salient design characteristics; specification number.
(c) Blank space for Government inspector's official stamping.
(d) National stock number.

3.11.3 Diagram of connections. A diagram of connections shall be attached to the inside of the cover of the welding machine or inside the welding control panel cover. Such diagram of connections shall be protected by a coating of transparent varnish or other suitable material.

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3.12 Workmanship.

3.12.1 General. Workmanship shall be in accordance with the requirements herein applicable to marking of parts and assemblies, wiring, welding, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part unsatisfactory for the purpose intended.

3.12.2 Threaded parts or devices. Screws, nuts, and bolts shall show no evidence of cross threading, mutilation, or detrimental or hazardous burrs.

3.12.2.1 Tightness. Screw-type fasteners shall be tight. The word tight means the screw shall be firmly secured and that there shall be no relative movement possible between the attached parts.

3.12.3 Wiring. Insulated wire shall be formed into cables wherever practicable. Wires and cables shall be positioned or protected to avoid contact with rough or irregular surfaces and sharp edges.

3.13 Technical manuals. When specified by the contract or order (see 6.2.1), technical manuals (see 6.2.3) shall be prepared in accordance with the data ordering document (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified in the detailed tests herein, the inspection and tests as specified in 4.6.2 shall be performed under the following conditions. Ambient conditions within the specified range need not be controlled:

(a) Temperature: 25 ± 10°C.
(b) Humidity: 0 to 95 percent R.H.
(c) Power input: Nominal power of 3.4.3.
(d) Attitude: Normal operating position.

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4.4 First article inspection. First article inspection shall consist of the examinations and tests as specified in table IV. The examinations and tests shall be performed on one welding machine in the order listed. The contractor shall prepare a first article inspection report in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in table IV. The examinations and tests shall be conducted in the order listed for each group. The contractor shall prepare test reports in accordance with the data ordering documents included in the contract or order (see 6.2.2).

TABLE IV. First article and quality conformance inspections.

Examination or test	Requirement paragraph	Examination or test paragraph	First article	Quality conformance inspection		
				Group A	Group B	Group C
Mechanical and visual	3.12	4.6.2.1	X	X		
Insulation resistance	3.4.5.1	4.6.2.2	X	X		
Power input interface	3.7.1	4.6.2.3	X			
Practical welding test	3.7.2	4.6.2.4	X			
Output current range	3.7.3.2.1	4.6.2.5	X		X	X
Output current adjustment and stability	3.7.3.2.3 3.7.3.2.4	4.6.2.6	X		X	X
Temperature rise	3.7.6	4.6.2.7	X			X
Dielectric tests	3.4.5.2	4.6.2.8	X	X		
Efficiency	3.7.7	4.6.2.9	X			X
Shock tests	3.8.4	4.6.2.10	X			
Vibration	3.8.3	4.6.2.11	X			

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TABLE IV. First article and quality conformance inspections. - Continued

Examination or test	Requirement paragraph	Examination or test paragraph	First article	Quality conformance inspection		
				Group A	Group B	Group C
Inclination	3.8.5	4.6.2.12	X			
Humidity	3.8.2	4.6.2.13	X			
Weight	3.9	4.6.2.14	X			X
Reliability	3.5	4.6.2.15	X			
Maintainability	3.6	4.6.2.16	X			
Electromagnetic compatibility	3.7.8	4.6.2.17	X			X

4.5.1 Sampling for quality conformance inspection.

4.5.1.1 Inspection lot. Welding machines conforming to this specification, identified for inspection at one time shall be considered a lot for quality conformance inspection.

4.5.1.2 Sample size. Sample size shall be as required in 4.5.1.2.1 through 4.5.1.2.3.

4.5.1.2.1 Group A. Each welding machine shall be subjected to the group A inspection.

4.5.1.2.2 Group B. Sample welding machines shall be selected from each lot in accordance with MIL-STD-105, general inspection level II with acceptance quality level (AQL) of 1.5 percent and subjected to the group B tests.

4.5.1.2.3 Group C. Sample welding machines shall be selected from each lot in accordance with table V and subjected to the group C tests.

TABLE V. Sampling for group C tests.

Number of machines in lot	Number of units to be tested
1 to 200	1
201 to 500	2
501 and over	3

4.5.2 Classification of defects. Failure to satisfy any test specified in 4.6.2 shall be counted as a defect. All defects are classified as major, as specified in MIL-STD-105.

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4.5.3 Acceptability of lots. Acceptability of a given lot shall be determined as follows:

Group A: in accordance with table VI.
Group B: in accordance with MIL-STD-105.
Group C: in accordance with table VI.

TABLE VI. Lot acceptability.

Lot size	Maximum acceptable number of defectives	
	Group A	Group C
40 and under	0	0
41 and 300	1	0
301 and over	2	0

4.5.4 Rejected lots. Resubmittal of rejected lots shall be in accordance with MIL-STD-105.

4.6 Methods of inspection.

4.6.1 Test equipment. Test equipment shall be as specified in MIL-E-917, except that the shock machine shall be in accordance with MIL-S-901.

4.6.2 Inspections.

4.6.2.1 Mechanical and visual. The welding machine shall be subjected to mechanical and visual inspections to determine that the workmanship is uniform in quality and free of defects that will adversely affect life, serviceability or appearance, and for conformance to the design requirements of this specification.

4.6.2.2 Insulation resistance tests (cold). Prior to application of the test voltage, the circuits shall be thoroughly discharged. Circuits of equal voltage above ground shall be connected together. Circuits or groups of circuits of different voltages above ground shall be separated. Separate measurements shall be made on circuits which are of different voltage above ground. Insulation resistance shall be measured with an insulation resistance meter or test bridge having a direct current test voltage of approximately 500 volts and a range of 0.1 to 100 megohms. The time of test voltage application shall be not less than 60 seconds. All readings, the applicable circuit for each reading, and the temperature at which these readings were taken shall be recorded in the test report. Any components having a test voltage rating less than the test voltage shall be removed or protected from the test voltage during this test. These components shall be tested at the test voltage proper for that component.

4.6.2.3 Power input interface tests.

4.6.2.3.1 Current waveform test. The harmonic (between input line frequency and 20 kHz) content of current waveforms in the welding machine power input lines shall be determined for each operating mode (see 3.7.1.1).

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4.6.2.3.1.1 Apparatus. The following apparatus is recommended for this test:

(a) Power source.
(b) Voltmeters (true rms) - plus or minus 1/2 percent accuracy.
(c) Frequency meter - plus or minus 1/2 Hz accuracy.
(d) Ammeters.
(e) Wide-band current probe or shunt.
(f) Potential transformers as required.
(g) Harmonic analyzer/electromagnetic interference meter.

4.6.2.3.1.2 Procedure. The welding machine shall be energized in accordance with the equipment procedure and operated in a normal mode. The power source voltage and frequency shall be within the user tolerance envelopes. The power input current harmonics shall be determined for each operating mode. The current harmonics shall be measured by means of a wide-band current probe or shunt in each power input line. The fundamental and harmonics of each line current shall be measured by means of a harmonic analyzer or electromagnetic interference meter. The current harmonics shall be determined for each normal operating mode.

4.6.2.3.2 Voltage and frequency tolerance test. Voltage and frequency tolerance test shall be used to evaluate the performance (thermal and operational) of the welding machine under the voltage and frequency tolerance specified in 3.7.1.2.

4.6.2.3.2.1 Apparatus. The following apparatus is recommended for performing the voltage and frequency tolerance test:

(a) Power source of required capacity and range of voltage and frequency adjustments. A power source capable of having an independently programmable voltage and frequency output is recommended.
(b) Voltmeters (true rms) - plus or minus 1/2 percent accuracy.
(c) Frequency meter - plus or minus 1/2 Hz accuracy.
(d) Temperature meter - plus or minus 3 degrees accuracy.

4.6.2.3.2.2 Procedure. The welding machine shall be operated in one of the normal operating modes within the user frequency and voltage tolerance in 3.7.1.2 until the equipment temperature has stabilized. The power input voltage and frequency shall then be varied in accordance with 3.7.1.2, and the welding machine shall be operated at each voltage and frequency combinations until the equipment temperature has stabilized and for a period of 1 hour, thereafter. This test shall be repeated for each mode of equipment operation. Frequency, voltage and internal equipment temperatures shall be measured and recorded as required until temperatures have stabilized and at 30 minute intervals, thereafter.

4.6.2.3.3 Voltage and frequency tolerance and recovery test. The voltage and frequency tolerance and recovery test shall be used to evaluate the welding machine under the conditions specified in 3.7.1.3.

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4.6.2.3.3.1 Apparatus. The following apparatus is recommended for performing the voltage and frequency transient tolerance and recovery test:

(a) Power source.
(b) Voltmeters (true rms) - plus or minus 1/2 percent accuracy.
(c) Frequency meter - plus or minus 1/2 Hz accuracy.
(d) Recording oscillograph or storage oscilloscope having 500 kHz response.
(e) Current and potential transformers and probes as required.

4.6.2.3.3.2 Procedure. The welding machine to be tested shall be operated in a normal operating mode within the user voltage and frequency tolerance band until the equipment temperatures have stabilized. The power input voltage and frequency shall then be suddenly changed to the upper limits specified in 3.7.1.3 within 0.1 second, simultaneously. Return the voltage and frequency to the user tolerance band within the recovery time specified in 3.7.1.3. Input voltages, frequencies and input line currents shall be recorded prior to the initiation of the voltage and frequency transient and until the transient is completed. Initial condition of voltage, frequency and line current shall be measured and recorded prior to the start of each test. Repeat the test for the lower limit of voltage and frequency specified in 3.7.1.3. Request the test for each normal mode of operation.

4.6.2.3.4 Voltage spike test.

4.6.2.3.4.1 Apparatus. The following apparatus is recommended for conducting the voltage spike test:

(a) The power source shall be capable of withstanding the voltage spike test.
(b) Solar Spike Generator, Model 7399-1 (modified) or equivalent.
(c) Oscilloscope with Tektronix P6103A Probe or equivalent.
(d) Pearson wide-band current transformer or equivalent.
(e) High value capacitors, G.E. Company, 28F Series or equivalent.

4.6.2.3.4.2 Procedure. The welding machine shall be disconnected from the power source and the voltage spike waveshape shown on figure 1 shall be superimposed on the power source voltage. The voltage exhibited on the power system shall have a peak value of 2400 to 2500 volts. The voltage spike shall be developed between any two power lines and from each power line to ground. After establishing the voltage spike on the power system, the welding machine shall be reconnected and the voltage spike shall be applied. The test shall be repeated for each pair of power lines and from each line to ground.

4.6.2.3.4.3 Figure 2 is recommended for use in the application of line-to-line and line-to-ground voltage spikes on the welding machine requiring 60 Hz, three-phase, 440 volts alternating current power. Capacitors of equal value are connected from each power line to ground (or chassis) (see figure 2). The line-to-ground capacitors represent a lumped equivalent of the distributed and connected line-to-ground capacitance present in 60 Hz power systems. Line-to-ground capacitance measured aboard various ships has ranged from 1.5 microfarad (μF) to 86.8 μF. For purposes of this voltage spike test, an initial value of 20 μF should be used for the value of C_g on figure 2.

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4.6.2.3.4.4 Detail procedure. The voltage spike generator (VSG) (Solar Model 7399-1 or equal) output is connected between a power line and line-to-ground capacitor as shown on figure 2. Connecting the VSG in this manner results in the simultaneous addition of a voltage spike to a line-to-ground voltage waveform and to two-phase voltages when the ignitron fires. In this test circuit, line-to-line capacitors cannot be connected between phases because suppression of the transient would occur. Due to the design of the VSG, the voltage spike is synchronized to the VSG input power when operating in the 60 Hz SYNC MODE. Therefore, for 60 Hz applications, the VSG and the welding machine shall be supplied by the same power source to permit synchronization of the voltage spike to the welding machine input voltage. The VSG is rated for 115 volts, so a "step-down" transformer is needed for a 440 volts power source when synchronization is desired. Since voltage spikes are applied to the power source and other equipment operating on the line, as well as the welding machine, a dedicated power source that can tolerate high voltage spikes should be used to power the welding machine. To prevent damage to the VSG and other test equipment, synchronization shall be obtained from the phase in which no transient appears. Table VII shows which phase should be used.

TABLE VII. Synchronization phase for voltage spike test.

Line-to-ground VSG connection	Voltages displaying transient	Synchronization phase
Line A	V_{ag}, V_{ab}, V_{ca}	Ø BC
Line B	V_{bg}, V_{ab}, V_{bc}	Ø CA
Line C	V_{cg}, V_{bc}, V_{ca}	Ø AB

Peak voltage amplitudes exceeding 2500 volts are readily obtained in this test. Therefore, the following steps should be taken to limit the peak voltage (voltage spike plus line-to-ground voltage) to 2500 volts.

(a) First, with the welding machine disconnected, select the desired location of the voltage spike on the power source voltage waveform using the synchronizing procedure specified in the Model 7399-1 Spike Generator, Solar Electronics' instruction book or equal.

(b) With the welding machine disconnected, application of low amplitude voltage spikes can commence; and the voltage spike amplitude is gradually increased until 2500 volts peak amplitude is observed on any power source voltage. The welding machine can then be reconnected and voltage spike tests conducted.

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(c) To change the location of the voltage spike on the power source voltage waveform, disconnect the welding machine and repeat the procedure in (a) and (b) above. In the test circuit of figure 2 the power source and line-to-ground capacitance are in parallel with the internal load of the VSG which consists of an air core inductor, L_p, and a resistor, R_p, connected in parallel. Should the source impedance or the line-to-ground capacitance be low in value, a considerable decrease in pulse width occurs. Restoration of the pulse width to the specified value will entail re-selection of line-to-ground capacitance values and modification of the VSG discharge circuit. The following simple modifications to the Solar Model 7399-1 VSG or equal should be implemented if the required pulse width is not obtained when first establishing the voltage spike on the power system with the welding machine disconnected:

(d) Disconnect inductor L_p

(e) Add a power resistor, R_s, in series with the 0.667 ohm resistor, presently installed in the VSG as shown on figure 2.

Initial component values for R_s and each line-to-ground capacitor, C_g, are 1.0 ohms and 20 µF, respectively. The voltage spike width will increase or decrease as R_s or C_g is increased or decreased, respectively. R_s should be a power rib or ribwound resistor. The resistor ribs should be spaced wide enough to prevent arcing across the ribs during the discharge. The required power rating of R_s will depend on the power source voltage and on the component values selected for R_s and C_g. For example, when the power source voltage is 450 volts a.c., the line-to-ground voltage is 260 volts a.c. When C_g is 20 µF, the maximum line-to-ground current through R_s is approximately 2 amperes. Therefore, when R_s is 10 ohms, a resistor power rating of 40 watts is required; and when R_s is 25 ohms, a power rating of 100 watts is required. Each line-to-ground capacitor should be rated for a peak voltage amplitude of approximately 1600 volts. When a high value of line-to-ground capacitance is required to develop the specified pulse width, high line-to-ground currents will flow. These currents could become of sufficient magnitude to require an unreasonable power rating for resistor R_s and cause the power rating of R_p (75 watts) to be exceeded. Increasing the value of discharge capacitance, C_p, in the VSG will overcome this difficulty. This VSG modification will increase the voltage spike width without having to increase the line-to-ground capacitance. The Solar spike generator can be modified as follows:

(f) Increase of VSG discharge circuit resistance.

(1) Open VSG cabinet.

(2) Disconnect wire from upper output jack and insulate wire lug with rubber boot or other means.

(3) Disconnect 0.667 ohms discharge resistor, R_p (three 2-ohm resistors wired in parallel), from 50 microhenry (µH) inductor, L_p. The black wires from the three VSG discharge capacitors are soldered to the wire connected to the three 2-ohm resistors.

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(4) Disconnect the inductor and discharge resistor from the ignitron mounting assembly. Insulate the disconnected inductor lug. The inductor is now out of the discharge circuit.

(5) A resistor of the desired value is added (the added resistor (R_s) should be a power rib or ribwound resistor with wide rib spacing to prevent arcing during the discharge) to the discharging circuit as follows:

a. Connect a wire from one end of R_s to the lower wire from the discharge resistor. Insulate the connection. Attach two wires to the other end of R_s. Connect one wire to the upper output jack and the other wire to the ignitron mounting assembly. Resistor R_s is now in series with the discharge resistor. Heavy gage wire with multiple strands shall be used to make all connections.

(g) Increase of VSG discharge circuit capacitance.

(1) Remove the brass nut which fastens the semicircular brass plate to the ignitron anode. The red wires from the positive terminals of the VSG discharge capacitors are soldered to this plate. Connect the positive terminal of the additional discharge capacitor to the ignitron anode and replace the brass nut.

(2) Connect the negative terminal of the additional capacitor to the lug of the upper wire from the three 2-ohm resistors connected in parallel. The black wires from the VSG discharge capacitors are soldered to this wire.

(3) Any additional discharge capacitor shall have a rating of not less than 3000 volts d.c.

4.6.2.4 Practical welding tests. Practical welding tests to determine suitability of operation shall be conducted by a qualified arc welding operator (qualified in accordance with MIL-STD-248) and shall include welding in all positions with MIL-7018 electrodes in accordance with MIL-E-22200/1 to determine compliance with 3.7.2 and 3.7.3.2.3.

4.6.2.5 Output current range tests. Output current range tests shall be conducted so that volt-ampere curves may be plotted for three settings; with the controls set for maximum, average (150 amperes) and minimum output. If more than one control is used these tests shall be conducted at the minimum, average, and maximum for each control. Readings shall be taken at points corresponding to no load, at short circuit and at least six points between no load and short circuit for each of the above referenced settings. Measurements taken shall be recorded.

4.6.2.6 Output current adjustment tests. Output current adjustment tests shall be conducted to show conformance with the current adjustment requirement as specified in 3.7.3.2.3.

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4.6.2.7 Temperature rise tests. The temperature rise tests shall be made on the completely assembled welding machine.

4.6.2.7.1 Conditions of tests. The temperature rises shall be determined under the following conditions:

(a) Continuous operation of the equipment with controls set to obtain a load equal to rated load amperes at rated load volts on duty cycles of 6 minutes at rated load followed by 4 minutes at no load until all temperatures are ultimate.
(b) Continuous operation of the equipment with approximately 20 volts at the terminals and with maximum current obtainable on the lowest current setting under a duty cycle of 6 minutes at maximum current followed by 4 minutes at no load until all temperatures are ultimate.
(c) Ultimate temperature is defined as the condition where the component temperatures do not increase when at least three consecutive readings are taken at not less than 10 minute intervals.
(d) Any blower motors shall be in continuous operation during temperature rise tests.

4.6.2.7.2 Measurement of ambient temperature. The ambient room temperature shall be measured by means of two or more thermometers placed at different points at a distance of 1 to 2 meters (3 to 6 feet) from the welding machine. The room temperature thermometers shall be inserted in oil filled cups of not less than 25 mm (1 inch) external diameter and 50 mm (2 inches) high, and shall be protected from drafts and from abnormal heat radiation. The value of the ambient temperature for the temperature rise tests shall be the mean of the readings of the thermometers. The conditions of the testing room shall be such that the room temperature variation does not exceed 10°C.

4.6.2.7.2.1 Temperature test data. During each temperature test the following data shall be taken at the end of the 6 minute load period after the welding machine has leveled off in temperature.

(a) Room ambient temperature.
(b) Temperatures of transformer and reactor windings.
(c) Operating temperature of the solid state power devices.
(d) Temperature of air being expelled from the equipment.

4.6.2.7.2.2 Methods of measuring temperature rises. Temperature rises shall be measured by method 1, 2 or 3 of MIL-E-917.

4.6.2.8 Dielectric tests. The rms test voltage shall be 1000 volts plus twice the rated voltage of the circuit under test. The frequency shall be 60 Hz and the wave shape essentially sinusoidal. The test voltage shall be applied continuously for a period of 1 minute. As an alternative to the 1 minute test, a test voltage which is 20 percent higher than the 1 minute test voltage applied for 1 second is acceptable for group A tests.

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4.6.2.9 Efficiency test. The input power shall be measured at the input terminals of the welding machine at rated voltage and frequency. The output shall be measured at the welding terminals. Measurements shall be made when the welding machine is delivering load amperes corresponding to 125, 100, 75, 50 and 25 percent of rated load amperes at a load voltage as determined by the formula specified in 3.7.3.2.2. The efficiency measurements shall be made immediately after the temperature tests at rated voltage and current while the welding machine is at approximately normal operating temperature. The percent efficiency shall be 100 times the measured output divided by the measured input.

4.6.2.10 Shock tests. The high impact shock tests shall be conducted in accordance with the grade A, type A requirements of MIL-S-901. Tests on the welding machine shall be conducted on the light weight shock machine. The welding machine shall be tested, on alternate blows, in each of two conditions, running and standing. During the running condition, the welding machine shall deliver rated voltage and current. During tests in the standing condition, the welding machine input terminals shall be connected to 440 volt, 3 phase, 60 Hz power with the switch or circuit breaker on the machine in the "OFF" position.

4.6.2.10.1 Inspection after shock test. The inspection after shock test shall include the following:

(a) Upon completion of the shock test, the welding machine shall be carefully inspected and tested to determine the extent of damage to the electrical and mechanical parts.
(b) The operating characteristics of the welding machine as obtained by the tests specified in 4.6.2.6 shall be spot checked to determine that they have not changed by more than 10 percent.
(c) The welding machine shall be operated under load until the operating temperatures of parts correspond to the values obtained at the completion of the temperature tests specified in 4.6.2.7.1(a). After the above operating temperatures have been attained, the welding machine shall be disconnected from its power source and load, and a dielectric test shall be conducted in accordance with the requirements of 4.6.2.8, except that the applied voltage shall be 1000 volts for 1 minute.
(d) Following the check tests specified in 4.6.2.10.1(a) through (c), the welding machine shall be disassembled and inspected for damage. The extent of disassembly shall be conducted to a point where the condition of the welding machine and its parts can be easily observed.

4.6.2.10.2 Causes for rejection. The following shall be cause for rejection of the welding machine under the high impact shock test:

(a) Breakage of any part.
(b) Nonconformity with the tests specified in 4.6.2.10.1(a) through (d).
(c) Activation of the welding machine from the deenergized condition or deactivation of the welding machine from the energized condition from any shock blow.
(d) Damage to the extent that proper operation or control of the welding machine cannot be exercised.

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4.6.2.11 Vibration. Vibration tests shall be conducted in accordance with type I requirements of MIL-STD-167-1. The welding machine shall be at rated load during this test.

4.6.2.11.1 Upon completion of the vibration test, the welding machine shall be inspected and tested to determine the extent of any damage to the electrical and mechanical parts.

4.6.2.11.2 The operating characteristics of the welding machine as obtained by tests as specified in 4.6.2.6 shall be spot checked to determine that they have not changed by more than 10 percent.

4.6.2.11.3 Causes for rejection. The following shall be causes for rejection of the welding machine under the vibration test:

(a) Breakage of any part.
(b) Nonconformity with the tests specified in 4.6.2.11.2.
(c) Damage to the extent that proper operation or control of the welding machine cannot be exercised.

4.6.2.12 Inclination tests. Inclined operation tests shall be conducted with the welding machine in each of three coordinate positions. Under each of these positions the welding machine shall be run at no load and full load each for a period of not less than one-half hour. Inclined operation shall have no effect on performance of the welding machine.

4.6.2.13 Humidity test. Prior to conducting the humidity test insulation resistance of all circuits to ground and circuit to circuit shall be recorded. The welding machine shall be placed in a humidity chamber at a R.H. of 90 to 95 percent at 40 $\pm$ 20°C for a period of 96 hours. Upon completion of this period insulation resistance shall be measured and recorded. The equipment should then be energized and operated at full load for 1 hour to demonstrate compliance with 3.8.2.

4.6.2.14 Weight. The weight of the welding machine complete with all component parts except power cable and welding leads shall be taken and recorded.

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4.6.2.15 Reliability. Endurance tests shall be conducted on the welding machine as a measure of reliability. The test shall be 50 hours duration with the following operating cycle.

Time duration (hours)	Input voltage 3 phase average	Input frequency (Hz)	Load at 60 percent duty cycle (6 minutes on 4 minutes off)
20 hours	440	60	100 percent for 10 hours 30 percent for 10 hours
5 hours	462	60	100 percent for 2-1/2 hours 30 percent for 2-1/2 hours
5 hours	418	60	100 percent for 2-1/2 hours 30 percent for 2-1/2 hours
5 hours	462	58.2	100 percent for 2-1/2 hours 30 percent for 2-1/2 hours
5 hours	418	58.2	100 percent for 2-1/2 hours 30 percent for 2-1/2 hours
5 hours	462	61.8	100 percent for 2-1/2 hours 30 percent for 2-1/2 hours
5 hours	418	62.8	100 percent for 2-1/2 hours 30 percent for 2-1/2 hours

There shall be no evidence of maloperation or excessive heating during the test. During the test, the equipment shall perform the functions as specified in 3.7, under the conditions of input electric power, voltage and frequency variations in accordance with DOD-STD-1399, section 300, without a failure for the duration of the test period.

4.6.2.16 Maintainability prediction. When specified in the contract or order (see 6.2.1), a maintainability prediction shall be performed in accordance with MIL-HDBK-472 (see 6.2.2) to determine compliance with 3.6.

4.6.2.17 Electromagnetic tests. Electromagnetic compatibility shall be tested in accordance with MIL-STD-462 to determine compliance with 3.7.8.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation, packaging, packing and marking. The equipment shall be preserved, packaged, packed and marked level A, B or C as specified (see 6.2.1), in accordance with MIL-E-16298.

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5.2 Technical manuals. Technical manuals, which accompany shipments prepared for level A, shall be packaged in accordance with method IC-1 of MIL-P-116. Technical manuals shall not be placed within the sealed water vaporproof barrier material used to enclose equipments. The copies of the manual shall be packed in the shipping container housing the main unit and packing lists shall indicate which container includes the technical manual and shall indicate the approximate location therein. Technical manual, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B or C as specified (see 6.2.1).

6. NOTES

6.1 Intended use. The welding machine described by this specification is intended for use aboard various types and classes of ships and boats to provide direct current for weld operations using ship's service alternating current power. It will remain aboard the ship and must therefore function properly under the shipboard environment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) First article, when required (see 3.1).
(c) Input power supply cable, when required (see 3.4.3.1).
(d) Reliability prediction, when required (see 3.5.1).
(e) If the MTTR or the M-max repair time is other than specified (see 3.6)
(f) Number of technical manuals required (see 3.13).
(g) Maintainability prediction, when required (see 4.6.2.16).
(h) Levels of preservation, packaging, packing and marking (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.5.1	Reliability prediction and documentation of supporting data	DI-R-7095	------
3.13	Manual with supplementary data, technical, commercial equipment	DI-M-2050	Type I of MIL-M-15071
4.4	First article inspection report	DI-T-4902	------
4.5	Reports, test	DI-T-2072	10.1.b
4.6.2.16	Maintainability prediction data	DI-R-5189	------

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2.3 Technical manuals. The ordering data should require that two technical manuals be supplied with each equipment. The manuals should be the manufacturer's commercial manuals furnished in substantial covers to withstand shipboard usage. The manuals should include the following:

(a) Certification data (see 6.2.3.1).
(b) Assembly instructions.
(c) Set-up or installation instructions.
(d) Operation instructions.
(e) Troubleshooting instructions.
(f) Maintenance instructions.
(g) Repair instructions.
(h) List and detailed identification of all repair parts.

6.2.3.1 Certification data. In addition to the contents required by DOD-STD-100, the certification data should include the following:

(a) Navy contract or shipbuilder's order and item number.
(b) Equipment manufacturer's order number.
(c) Equipment master drawing number.

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(d) Equipment catalog numbers.
(e) Applicable specification.
(f) Classification.
(g) Current rating.

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Terms and definitions. Quality assurance terms and definitions shall be accordance with MIL-STD-109. The following definitions apply to the terms as used in this specification.

6.4.1 Arc voltage. Arc voltage refers to the voltage drop which exists across the arc during the welding operation. Arc voltage is dependent upon the size and type of electrode used, the current and the length of the arc. Arc voltage should be considered as generally equal to the value determined from the formula as specified in 3.7.3.2.2.

6.4.2 Current. Unless otherwise specified herein, current is the output load current of the equipment.

6.4.3 Duty cycle. Duty cycle identifies the percentage of load time. A 60 percent duty cycle is defined as repetitions duty cycles each consisting of an operation at rated output for 6 minutes followed by an operation at no load for 4 minutes.

6.4.4 Hazardous item. A hazardous item is any substance, mixture, material, component or equipment which may cause personal injury, property damage or environmental deterioration through transportation, use or disposal.

6.4.5 Maximum open-circuit voltage. The maximum open-circuit voltage is the open-circuit voltage at the output terminals of the equipment when the equipment is set for maximum output, operating at no load, and with rated input supply (see 3.7.4.2).

6.4.6 Setting. A setting is defined as a given position of the equipment controls.

6.4.7 Toxic material. Toxic material is a material or combination of materials which are capable of producing dangerous gases or other harmful products when exposed to normal ambient temperature or to high temperatures, including fire.

6.4.8 Voltage. Unless otherwise specified herein, voltage is the output terminal voltage of the equipment.

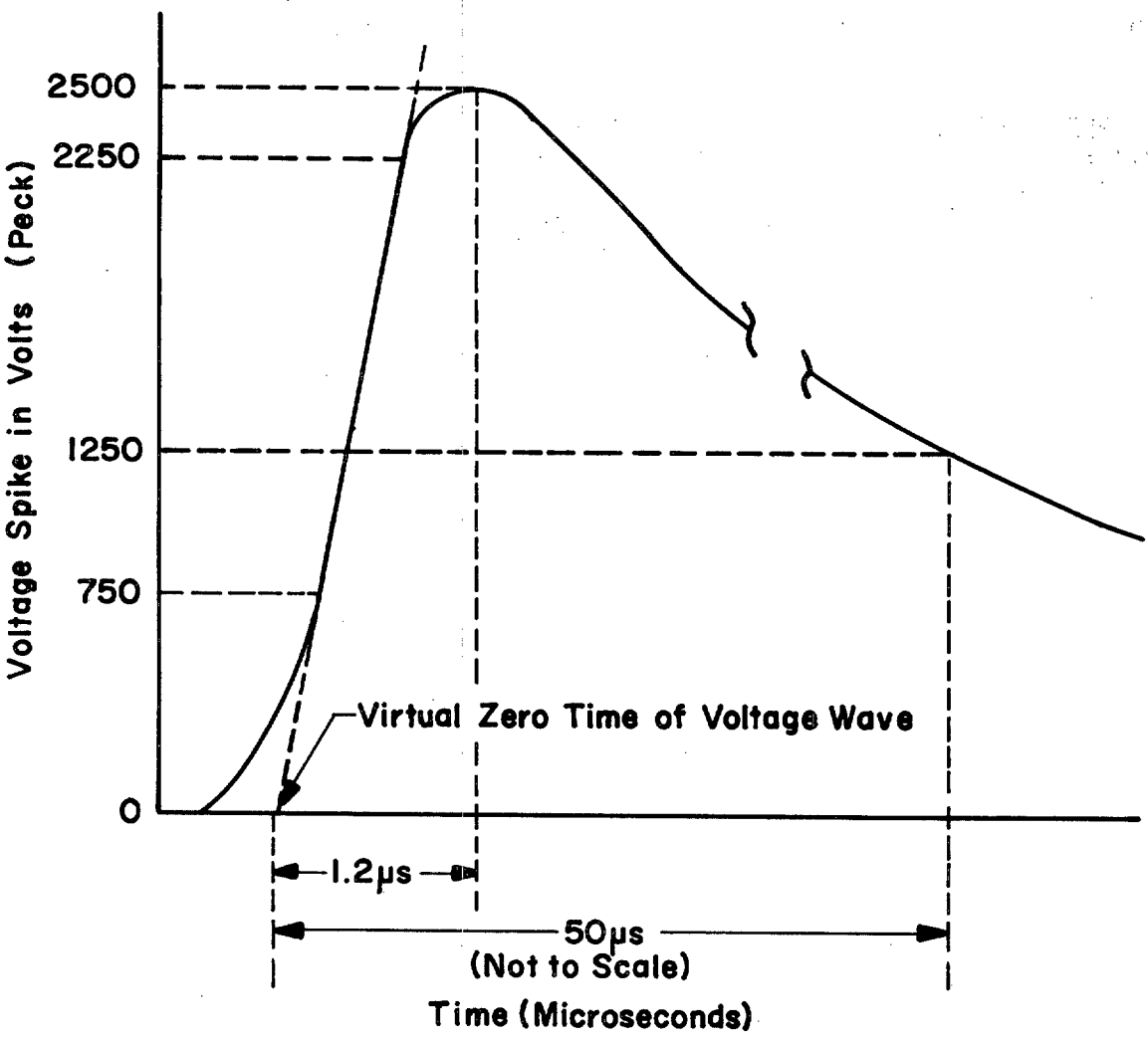
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6.4.9 Volt-ampere curve. The volt-ampere characteristic at any current setting is a plot of the terminal voltage versus output current for that setting when a connected load is varied to obtain an output current from zero to short-circuit current.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 3431-N160)

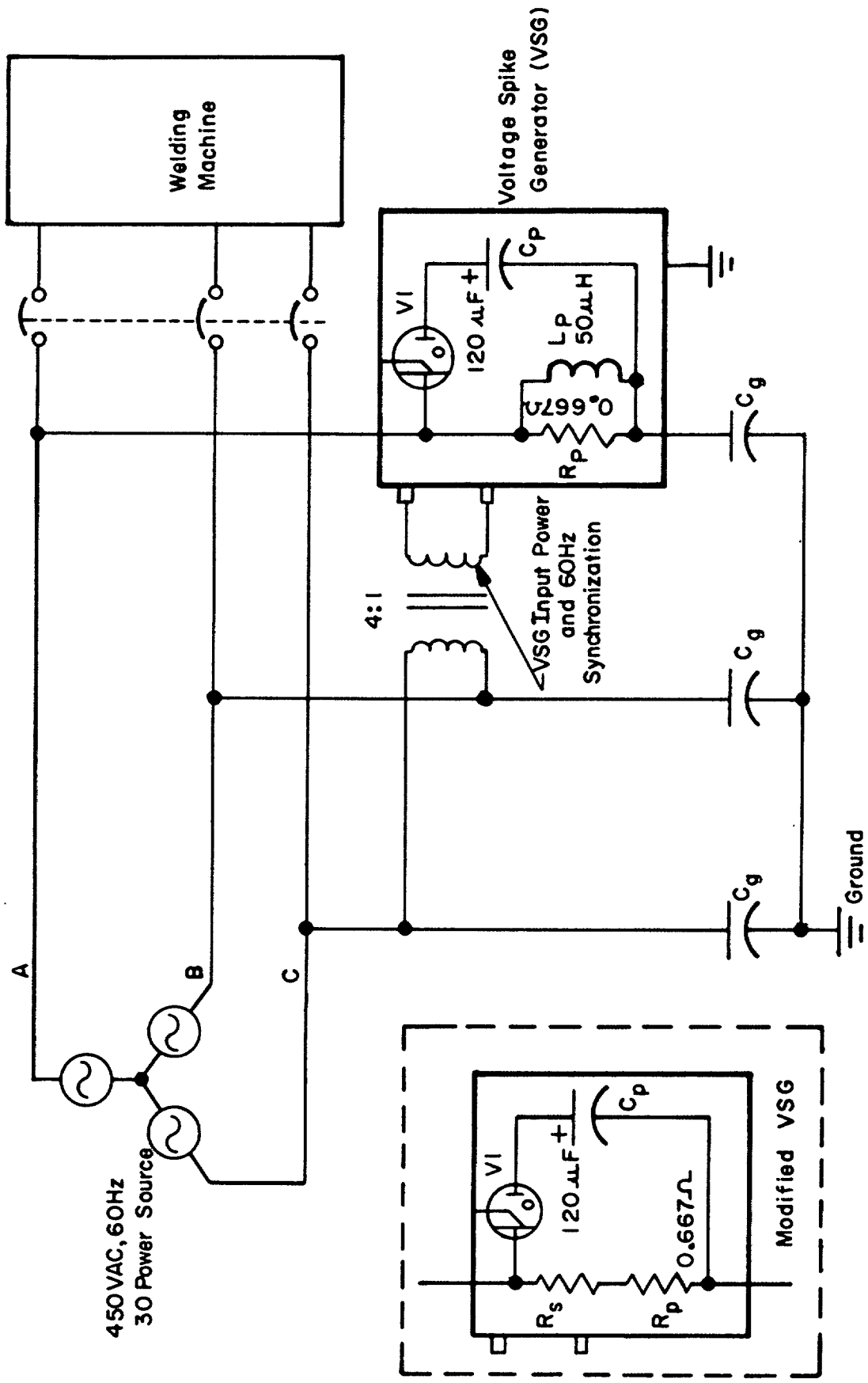
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SH 131121

FIGURE 1. Voltage spike impulse wave shape.

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SH 131122

FIGURE 2. Test circuit for application of line-to-line and line-to-ground high voltage spikes to welding machine supplied by three-phase, 60 Hz power.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side) Welding Machine, Arc, Direct-

1. DOCUMENT NUMBER	2. DOCUMENT TITLE Current 300 Ampere, Variable Voltage, Tr-
DOD-W-24644(SH)	former-Rectifier Type, Damage Control, Light Weight

3a. NAME OF SUBMITTING ORGANIZATION Portable (METRIC)	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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